Networks Decide for Us: Emergence of Adoption in Homophily-Imputed Social Topologies

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Background and Research Question

The diffusion of collective behaviors is frequently modeled as either individual utility maximization or a simple contagion process. Empirical reality, however, suggests a hybrid mechanism where individuals weigh both their intrinsic preferences and the influence of their peers [2, 8]. While recent models address this dual mechanism [7], computational literature often assumes synthetic network topologies, occasionally treating agents as cognition-free relay stations [1]. We address the methodological gap of ignoring deeply ingrained homophilic forces in the study of collective dynamics.

This research asks: Conditional on fixed individual propensities, how do empirically calibrated homophilic ties act as an autonomous causal force—effectively "deciding for us"—by simultaneously generating a structural premium in overall diffusion and dictating

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the fundamental dynamics of the social phase transition? Specifically, we explore how empirical homophily systematically amplifies the macroscopic scale of diffusion compared to randomized baselines, and how the disruption of this homophily shifts the system from structured, rolling sociological cascades to sudden, volatile tipping points.

Data and Imputation Pipeline

To build realistic network structures, we combine representative survey data with empirical homophily parameters. First, we use homophily targets estimated from core discussion networks [5], spanning dimensions of Blau space such as age, sex, education, race, and religion. Second, we imputed structure onto respondents from the American Trends Panel (ATP 2016) and a General Social Survey (GSS 2004) political module. These datasets provide individual propensities for 'openness to innovation' and 'collective action', respectively. The resulting topologies respect marginal demographic distributions while adhering to empirical homophily tendencies, producing realistic clustering that overcomes the limitations of standard theoretical network models [6, 3].

Methods: Hybrid Agent-Based Modeling

We analyze diffusion processes over these imputed topologies using a hybrid Agent-Based Model (ABM) simulated with the `netdiffuseR` package [9]. An agent i adopts ($a_i = 1$) if their intrinsic aversion (q_i) is overcome by the innovation's inherent utility level (Γ), or if their network threshold (τ_i) is overcome by the effective social exposure ($\tilde{E}_i$):

$$a_i = 1 \iff (q_i \leq \Gamma) \vee (\tau_i \leq \tilde{E}_i)$$

Crucially, the exposure $\tilde{E}_i$ only counts *influential* peers, weighted by social distance d_{ij} in Blau space and the system's Maximum Social Proximity (MSP), which sets an upper bound on how demographically different two interacting individuals can be to still exert influence:

$$\tilde{E}_i = \frac{\sum_{j \neq i} \mathbf{X}_{ij} \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}} \quad \text{where} \quad \tilde{a}_j = a_j \cdot \mathbb{I}\Big[U_{ij} \leq \sigma(MSP - d_{ij})\Big]$$

and σ is a standard logistic function.

Through approximately 4 million simulations, we explore the parameter space of intrinsic utility and social flexibility across both our plausible networks and degree-preserving randomization. To isolate the marginal effect of the randomized topology, we estimate the expected total adoption $\mathbb{E}(\Phi)$, where Φ represents the final proportion of adopters in the network, using Generalized Additive Models (GAM):

$$\text{logit}[\mathbb{E}(\Phi)] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{ER}}\mathbb{I}_{\text{Random}} + s(\Gamma, MSP)$$

where $\mathbf{Z}$ is a matrix of controls (e.g., threshold parameters, seeding strategies), $\mathbb{I}_{\text{Random}}$ isolates the random structural penalty, and $s(\Gamma, MSP)$ is a thin-plate regression spline capturing the non-linear interaction between intrinsic utility and social flexibility.

Preliminary Results

By comparing diffusion outcomes on plausible homophilic networks against random baselines, our regressions reveal how topology mathematically dictates both the volume and dynamics of collective behavior.

First, we identify a relevant structural premium. Randomizing the network structure depresses the odds of adoption by 57% ($\beta_{\text{ER}} = -0.850$, $p < 0.001$; see Table 1). Homophily acts as a structural catalyst: even if individuals possess high intrinsic thresholds reflecting an aversion to innovation or collective action, empirical clustering creates local echo chambers. These clusters allow social reinforcement to build safely and overcome local resistance.

Table 1: *GAM Regression Results: Effect of Network Structure and Seeding.*

	Rational Adoption	Social Adoption	Total Adoption
(Intercept)	0.428^{***}	-0.300^{***}	6.534^{***}
Thresholds			
τ_{mean}	-0.185^{***}	-5.130^{***}	-6.944^{***}
τ_{sd}	0.032	4.210^{***}	5.942^{***}
Seeding (Ref: Random)			
Central	-0.054^{***}	0.032^{***}	0.120^{***}
Closeness	-0.052^{***}	0.033^{***}	0.119^{***}
Marginal	0.027^{***}	-0.020^{***}	-0.067^{***}
Structure (Ref: Plausible)			
Random structure	-0.114^{***}	-0.426^{***}	$\mathbf{-0.850}^{***}$
Deviance Explained	96.9%	71.1%	82.2%

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Non-linear smooth terms $s(\Gamma, MSP)$ are highly significant ($p < 0.001$). Observations $(n) = 4,093,440$.

Second, homophily governs criticality dynamics. In empirical networks, social change unfolds as a predictable, rolling avalanche (Figure 1). Susceptibility diverges near the critical boundary MSP_c according to the universal scaling exponent γ [4]:

$$\chi = \text{Var}(\Phi) \propto |MSP - MSP_c|^{-\gamma}$$

We also observe Critical Slowing Down, where simulation relaxation times diverge as the system approaches the tipping point. Empirical data ($\gamma \approx 0.977 \approx 1$, see Table 2 and Figure 2) suggest that collective behavior obeys Mean-Field universal scaling laws. This

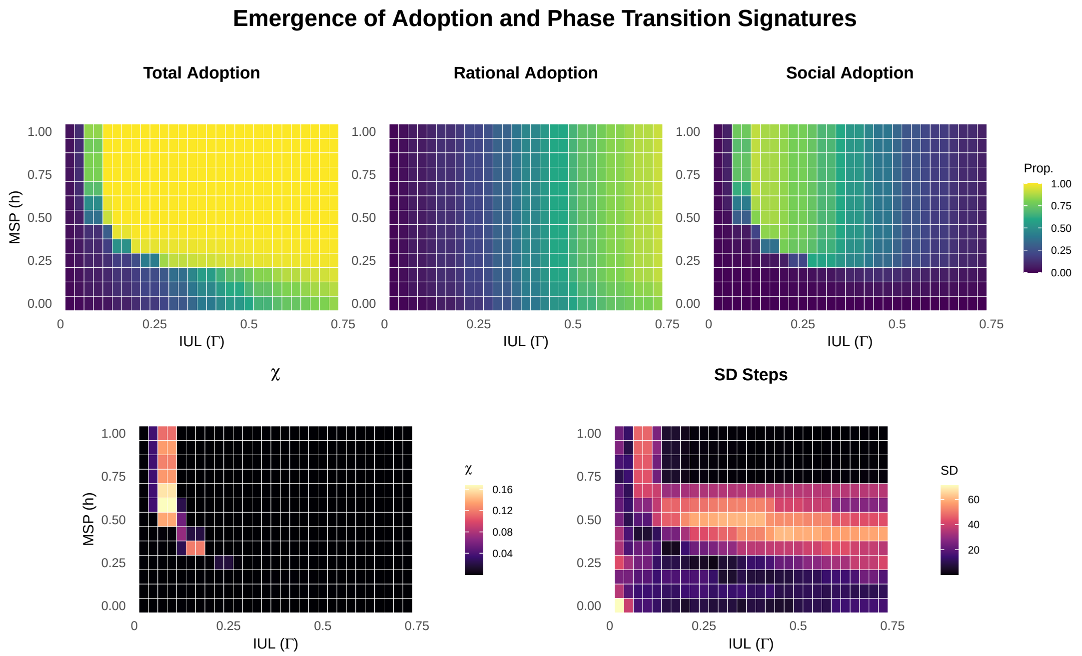

Figure 1: *Adoption Regimes and Phase Transition Signatures.* Top row: Average Total, Rational, and Social Adoption heatmaps. Bottom row (left): Susceptibility $\chi = \mathrm{Var}(\Phi)$ via the Fluctuation-Dissipation Theorem, with the phase boundary visible as a ridge of high variance. Bottom row (right): Standard Deviation of simulation relaxation time; its divergence is the signature of Critical Slowing Down. Simulations based on a GSS plausible network, $\tau_{sd} = 0.12$, and random seeding.

implies that social adoption operates as a scale-free process—much like a forest fire—where identical structural conditions can produce both localized clusters and massive, system-wide avalanches of adoption.

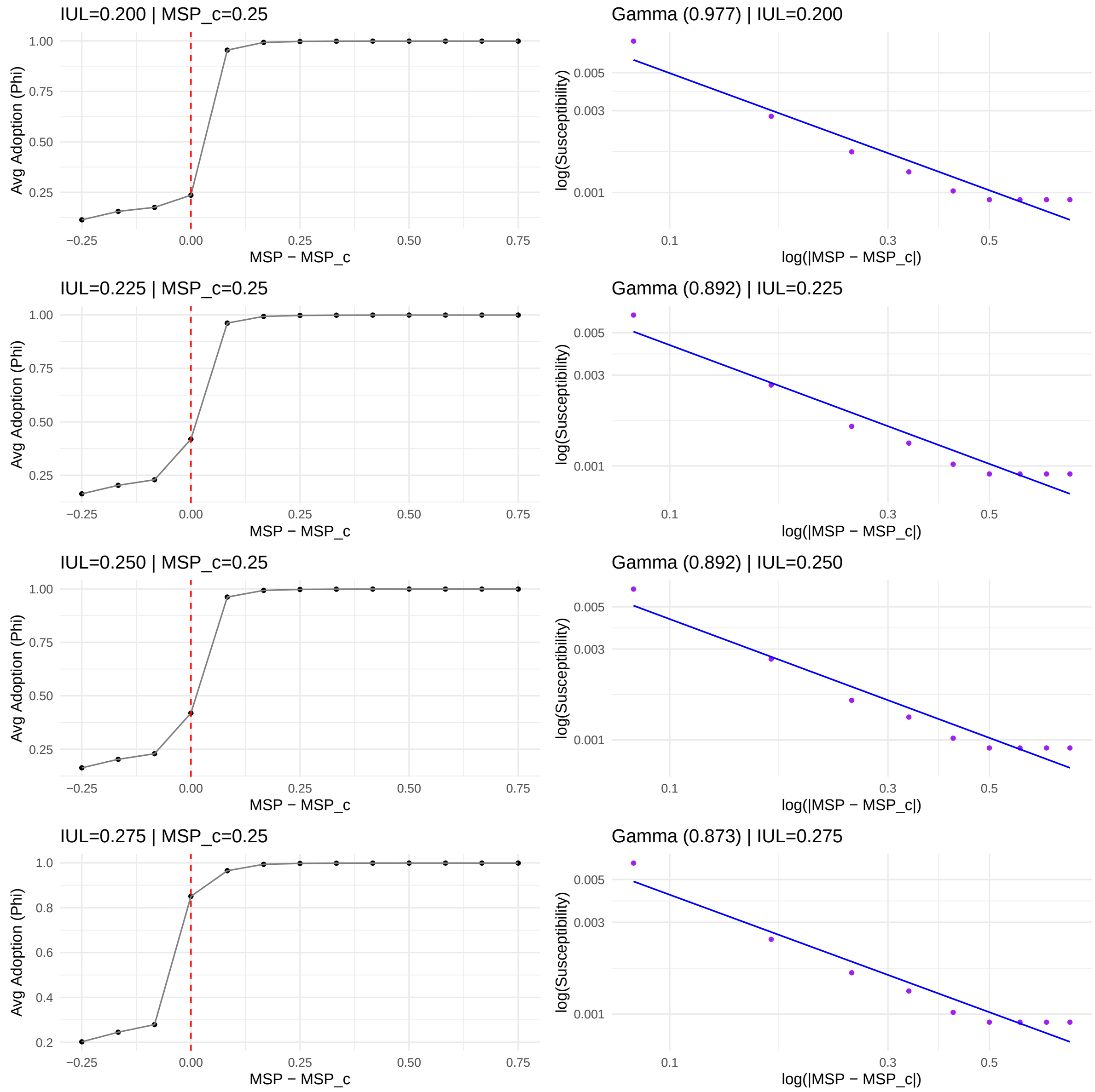

Figure 2: *Critical Exponents (Method 4).* Plots of the order parameter Φ against $MSP - MSP_c$ (left panels) and log-log plots of susceptibility χ against $|MSP - MSP_c|$ (right panels) for four IUL values along the Widom Line. Linear fits in log-log space confirm power-law scaling with exponent $\gamma \approx 0.977$, consistent with Mean-Field universality.

Ultimately, our results validate the premise that networks decide for us in a two-fold manner. The topological arrangement acts as a fundamental causal mechanism that not only systematically amplifies the volume of collective behavior (the structural premium) but also governs the underlying dynamics of social tipping points. By reshaping the critical thresholds, empirical homophily determines whether a society undergoes behavioral shifts

Table 2: *Estimated Susceptibility Exponents (γ) via Maximum Derivative Heuristic.*

IUL	**GSS** MSP_c	**GSS** γ	**GSS-ER** MSP_c	**GSS-ER** γ
0.200	0.250	0.977	0.416	3.709
0.225	0.250	0.892	0.416	3.667
0.250	0.250	0.892	0.416	3.667
0.275	0.166	1.539	0.333	4.666

through structured sociological cascades or sudden, unpredictable regime shifts.

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